

Pencil Pressure Safety Lab

Spread a force before it dents the target.

Win condition: Largest area protected with clear data.

THE SCIENCE YOU ARE USING

Pressure vs force. Pressure is force divided by area. The same push on a sharp point dents a surface, but spread over a wide area it does not. That is why a sharp pencil pierces and a flat board does not: same force, different area.

Spreading stress. Safety design often means spreading a load so no single spot takes too much. You design a spreader that protects foam from a standard load and show the protected area with data.

HOW TO BUILD AND RUN IT

- 1 Test the bare point.** Press a standard load through a pencil point into foam. Measure the dent.
- 2 Design a spreader.** Build a cardboard spreader that distributes the same load over more area.
- 3 Apply the standard load.** Press the same weight through your spreader onto fresh foam.
- 4 Measure protected area.** Compare the dent. Less damage over more area means lower pressure.
- 5 Redesign for more spread.** Improve the spreader to protect a larger area and re-test.

Safety: No skin tests. Use foam only. Allergy: None.

MATERIALS

Pencils	shared
Foam	per team
Cardboard	shared
Weights	shared
Graph paper	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Pressure spread	40
Explanation	25
Data quality	20
Redesign	15
Total	100